Network Threats and Mitigation & Physical Security and Risk

1. Create a table listing 5 common network threats (such as phishing, DDoS, man-in-the-middle, malware, and password attacks), and for each, write one real-world example from an app or service you use (like WhatsApp, Instagram, or Paytm).

ANS :

Network Threat	Description	Real-World Example
Phishing	Fake messages or websites used to steal user credentials.	A fake Instagram login page sent through email asking users to log in.
DDoS Attack	Attackers flood a server with traffic to make it unavailable.	Users unable to access BookMyShow during high-demand ticket sales due to traffic overload.
Man-in-the-Middle (MITM)	An attacker intercepts communication between two parties.	Using WhatsApp on an unsecured public Wi-Fi network where data could be intercepted.
Malware	Malicious software that damages systems or steals data.	Downloading a fake Paytm APK containing malware from an unofficial website.
Password Attack	Attackers try to guess or crack passwords.	Repeated login attempts on an Instagram account using common passwords.

2. Simulate a brute-force attack scenario by writing a simple Python script that tries to guess a 4-digit PIN code by checking all possible combinations against a preset value. Constraint: The script should print the number of attempts it took to guess the correct PIN.

ANS :

```
# Preset PIN
correct_pin = "1234"

attempts = 0

for pin in range(10000):
    attempts += 1
    guessed_pin = str(pin).zfill(4)
    if guessed_pin == correct_pin:
        print("PIN Found:", guessed_pin)
        print("Attempts Required:", attempts)
        break
```

- **OUTPUT**

- PIN Found: 1234
- Attempts Required: 1235

- **Explanation**

  - The script checks every possible 4-digit PIN from 0000 to 9999.
  - It stops when the correct PIN is found.
  - It displays the total number of attempts needed.

3. List 4 physical security risks that could affect a server room hosting a popular app like Zomato or BookMyShow, and suggest one mitigation step for each risk.

ANS :

Physical Security Risk	**Impact**	**Mitigation**
Unauthorized Access	Attackers may steal or damage servers.	Use biometric access control.
Fire Hazard	Servers and networking devices can be destroyed.	Install fire suppression systems and smoke detectors.
Power Failure	Services may become unavailable.	Use UPS and backup generators.
Theft of Equipment	Loss of sensitive data and hardware.	Install CCTV cameras and locked server racks.

4. Draw a simple floor plan (hand-drawn or digital) of a small office and mark at least 3 physical security measures you would implement to protect network equipment (such as CCTV, biometric access, or cable locks).

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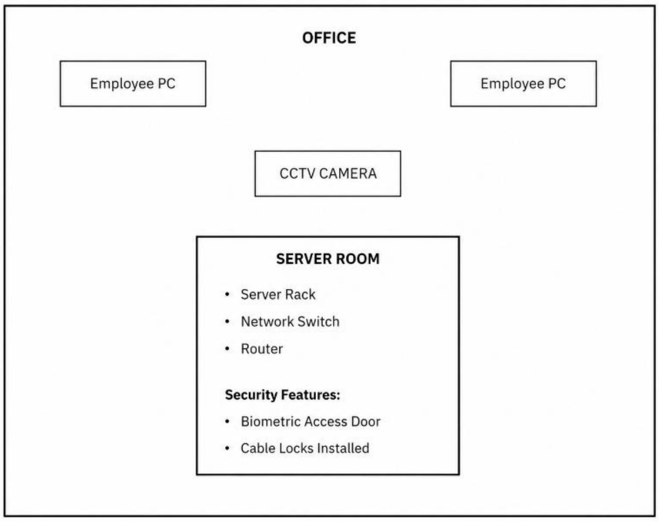
- Security Measures Used

1. CCTV Camera for monitoring.
2. Biometric Access Control for server room entry.
3. Cable Locks and Locked Server Rack to prevent theft.

5. Use ChatGPT or a similar AI tool to generate a checklist of 7 actions a network admin should take to mitigate risks from both network threats and physical security breaches, then review and edit the checklist to make it specific for a college computer lab environment.

ANS :

➢ Risk Mitigation Checklist



No.	Action
1	Install and regularly update antivirus software on all lab computers.
2	Enable Windows Firewall on every computer and block unnecessary ports.
3	Use strong passwords for student, faculty, and administrator accounts.
4	Restrict access to networking equipment using locked racks and cabinets.
5	Install CCTV cameras to monitor the computer lab and server area.
6	Perform regular backups of important academic and administrative data.
7	Keep operating systems, applications, and network devices updated with security patches.

➢ **Customized for College Computer Lab**

- Students should not have administrator privileges.
- USB storage devices should be restricted or scanned automatically.
- Router and switch rooms should remain locked.
- Faculty computers should use separate login credentials.
- Lab computers should automatically log off after inactivity.